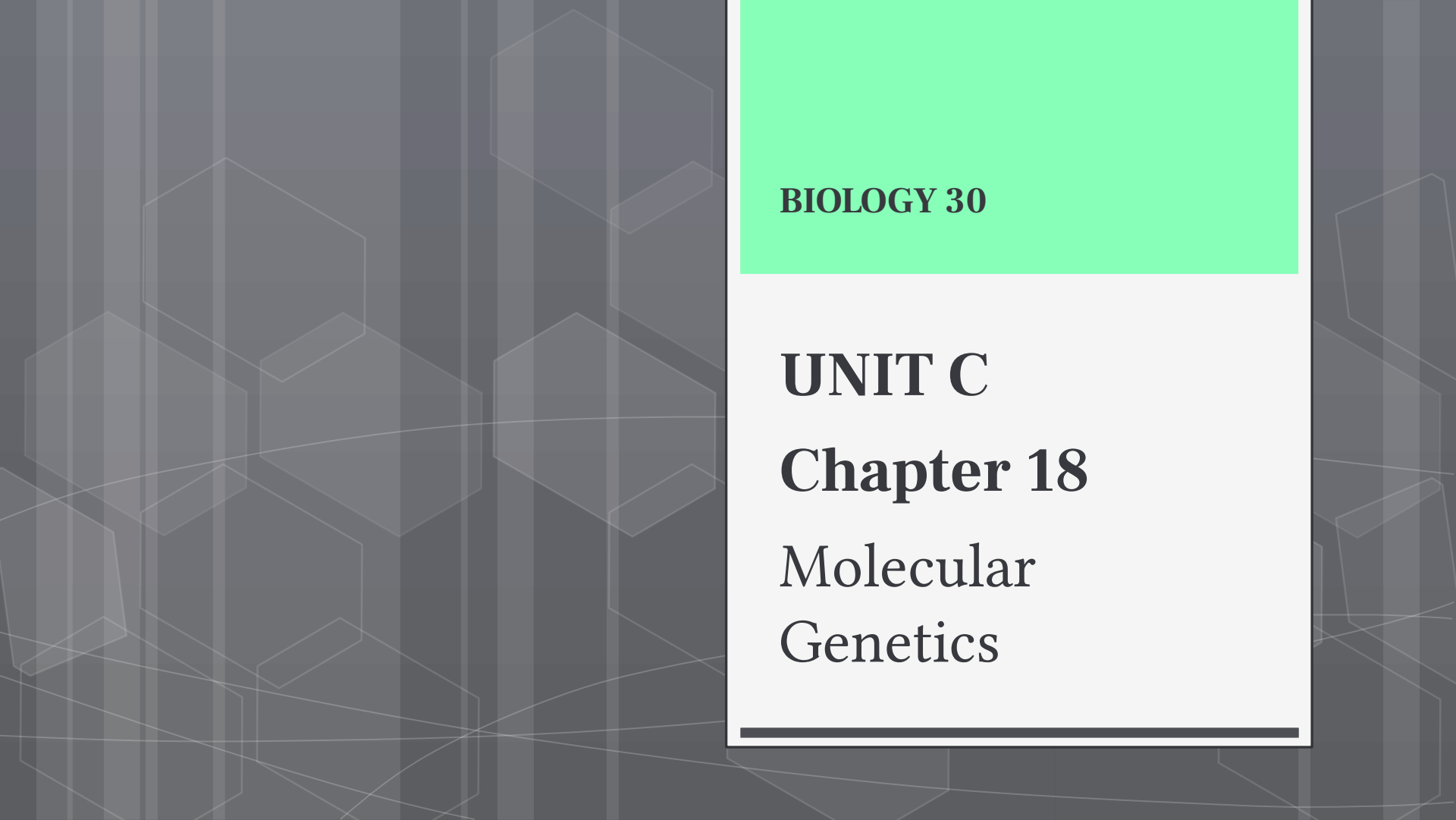


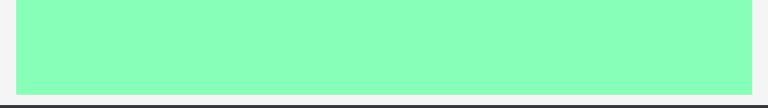


BIOLOGY 30

UNIT C

Chapter 18

Molecular
Genetics



TOPIC 1: DNA STRUCTURE & REPLICATION

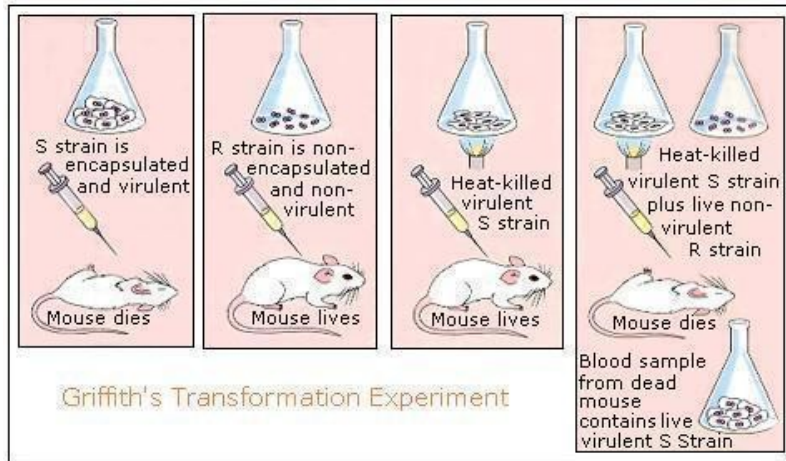
- Griffith's transformation experiment
- The Hershey-Chase experiment
- DNA structure
- Semiconservative replication
- Steps & enzymes involved in DNA replication
- The Meselson-Stahl experiment

Textbook pgs. 622-635

Note: We have to know some names of scientists and their contributions to our understanding of genetics and heredity. (i.e. testable)

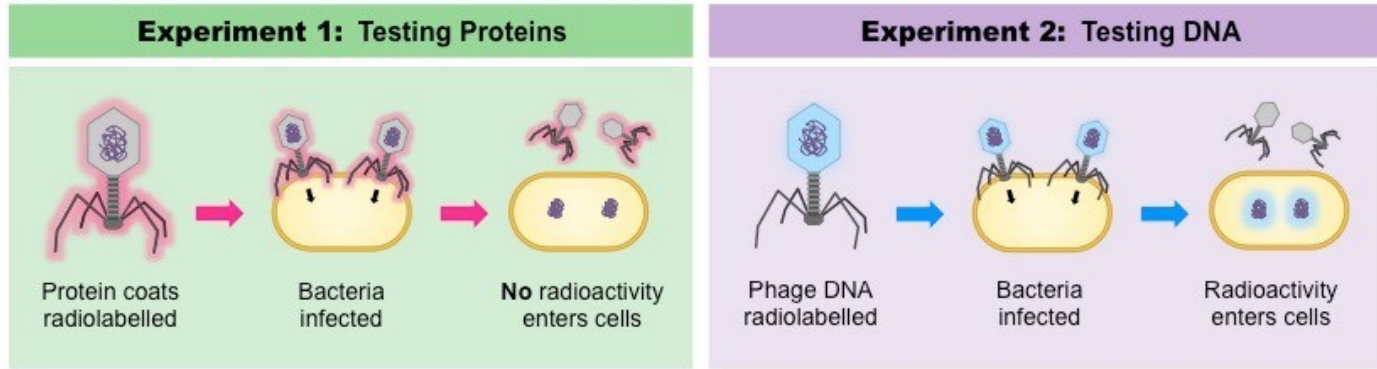
The “Transforming Principle”

In 1928, an experiment was conducted by Frederick **Griffith** which demonstrated that bacteria can be re-programmed through a process called **transformation**...



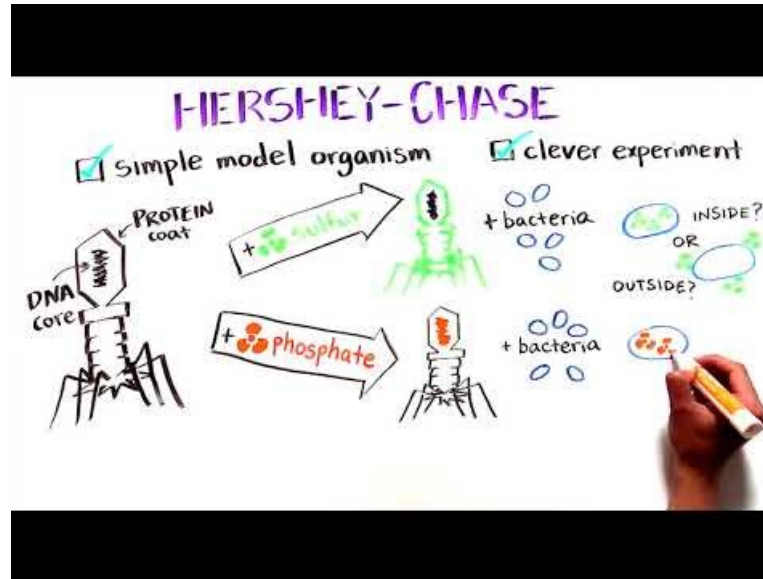
- Griffith's experiment suggested that something inside the cell must be responsible for its programming, and that this molecule could be transferred from cell to cell.
- The molecule that was responsible for the cell's programming (aka the hereditary material) was still up for debate...

A Case for DNA as the “Transforming Principle”



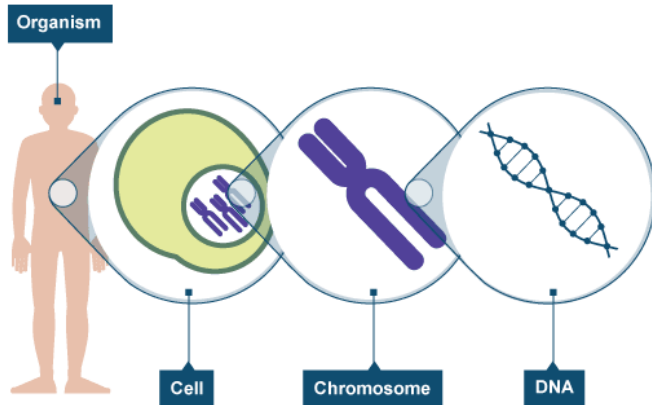
- o In 1952, Alfred **Hershey** and Martha **Chase** conducted a series of experiments to **prove that DNA**, and not protein, **was the hereditary material**.
- o This involved radio-labelling the proteins and DNA associated with a particular virus and allowing it to infect a bacterium to see which factor would be “injected” to transform the bacterium.

The Hershey-Chase Experiment:



DNA: The Hereditary Material

DNA is the nucleic acid molecule that governs the processes of heredity in all plant and animal cells.

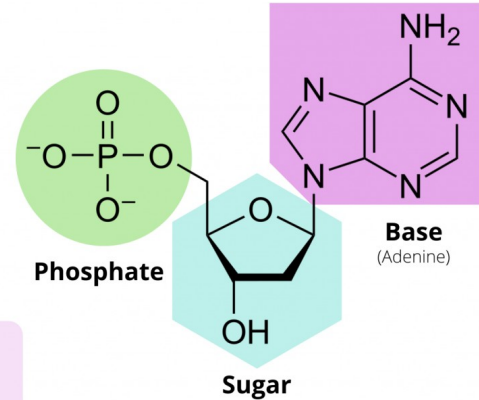


- Every cell in your body (with the exception of your gametes) contains the same DNA within its nucleus
- The way that DNA is **expressed**, however, differs from cell to cell
 - I.e. not every gene is turned on in every cell

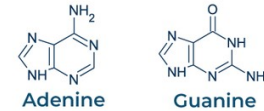
The DNA Nucleotide

- Dna is made up of long chains of individual units (monomers) called nucleotides. Each nucleotide consists of:
 - A five-carbon sugar
 - A phosphate group
 - One of four nitrogenous bases:
 - Adenine (A)
 - Guanine (G)
 - Cytosine (C)
 - Thymine (T)
- **Purines** have a double ring (adenine and guanine), whereas **pyrimidines** have a single ring (thymine and cytosine)

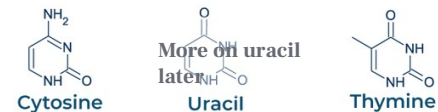
3 Parts of a Nucleotide



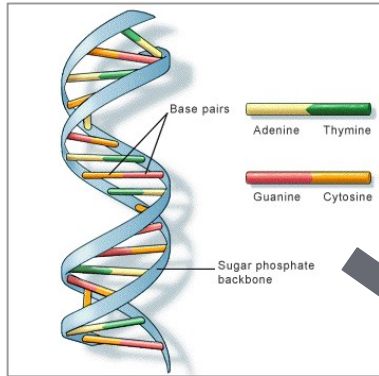
Purines



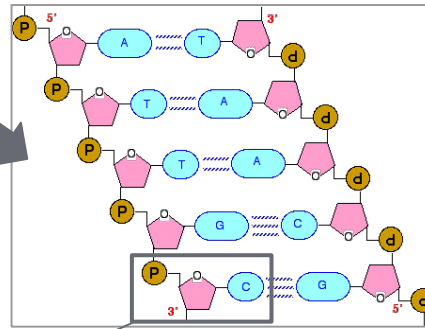
Pyrimidines



The Structure of DNA



DNA molecules form a *double helix*, with a sugar-phosphate backbone held together by *complementary* nitrogenous base pairs. The “rules” for each DNA molecule include...



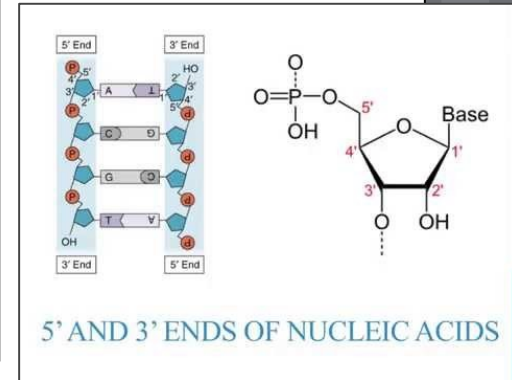
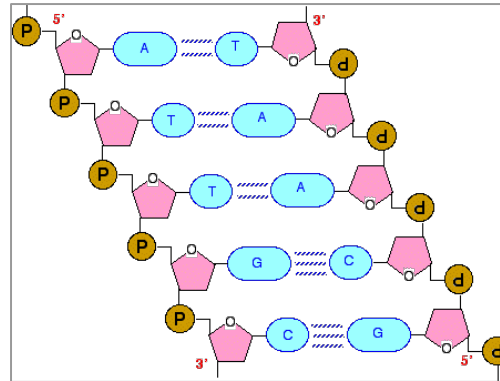
NUCLEOTIDE

1. **Hydrogen bonds** form between nitrogenous base pairs
2. Adenine (A) always pairs with Thymine (T)
3. Cytosine (C) always pairs with Guanine (G)
4. Each base is bonded to a sugar group.
5. Sugar groups are bonded to phosphates.
6. The grouping that exists between a phosphate molecule, a sugar molecule, and a single nitrogenous base is collectively referred to as a *nucleotide*

The Structure of DNA

DNA strands are also said to be *antiparallel*, meaning that they run in the opposite direction as one another:

- The 5' and 3' ends run in opposite directions
 - These numbers come from organic chemistry naming rules, and simply describe a particular carbon in the sugar, and thus the orientation of the nucleotide
- This property has important implications for DNA replication & protein synthesis



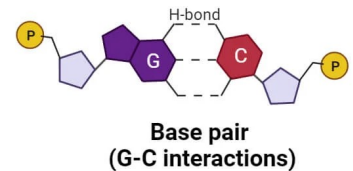
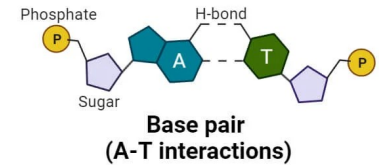
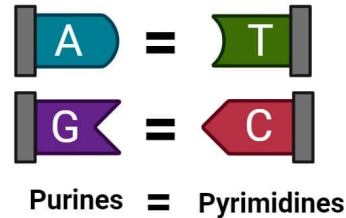
But how did we come to know so much about the structure of DNA?

In the 1940s, Erwin **Chargaff** figured out that the 4 different nitrogenous bases aren't present in equal amounts. They are present in varied, yet characteristic proportions where $A=T$ and $C=G$.

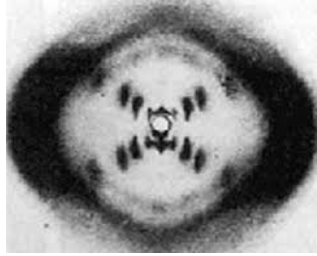
This became known as **Chargaff's Rule**.

- The AT:GC ratio is same for same species and different for different species
- The amount of A = the amount of T
- The amount of C = the amount of G
- $A+T$ does not necessarily equal $G+C$

Chargaff's Rule



Discovering the Structure of DNA:



- In the 1950s, British scientist **Rosalind Franklin** used **X-ray photography** to analyze the **structure of DNA**. Her observations provided crucial new information about the molecular structure of DNA.

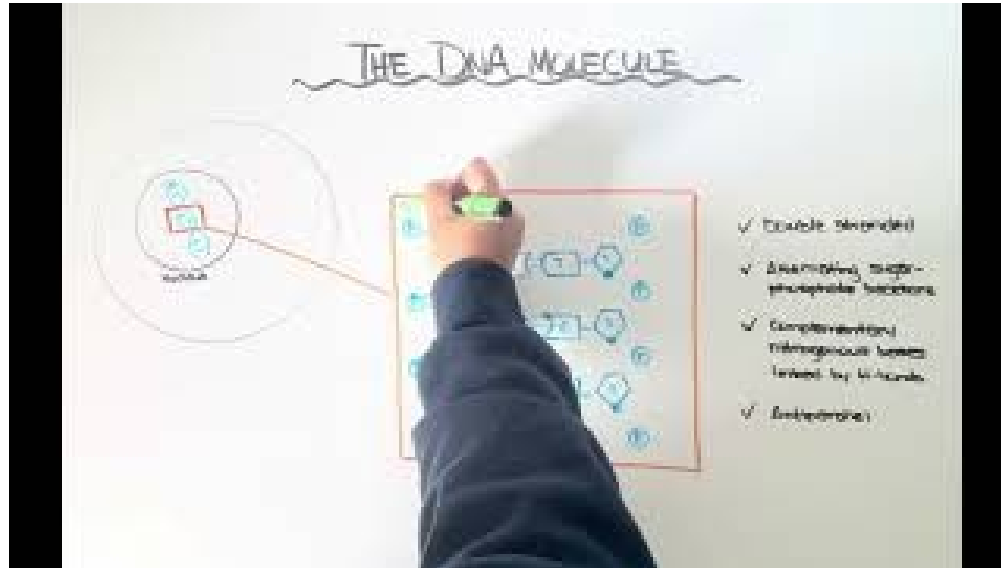
- In 1953, James **Watson** and Francis **Crick** were the first to produce the **structural model of DNA** that could account for all the evidence from Franklin's experiments as well as Chargaff's observations.

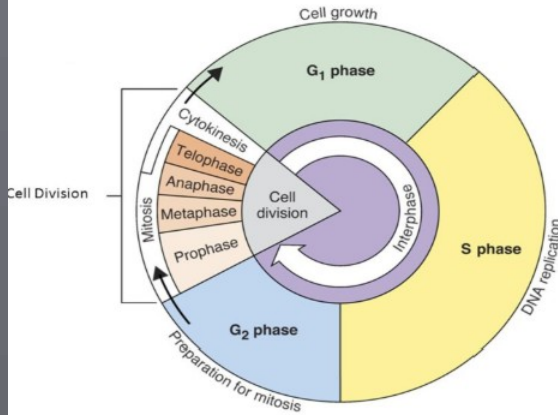


More reading if interested:

<https://www.sciencehistory.org/historical-profile/james-watson-francis-crick-maurice-wilkins-and-rosalind-franklin>

Re-cap: The DNA Molecule

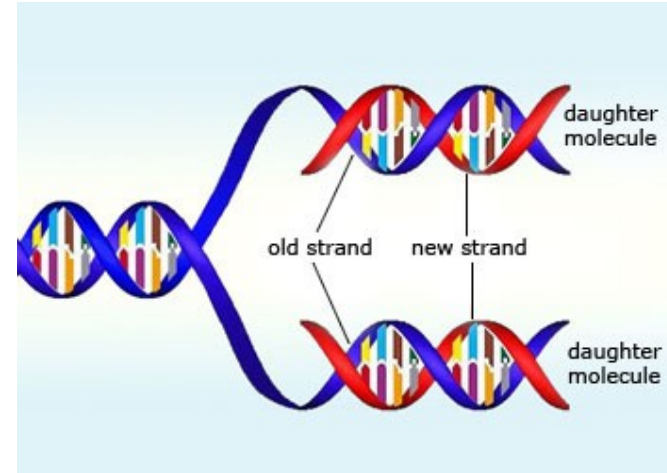




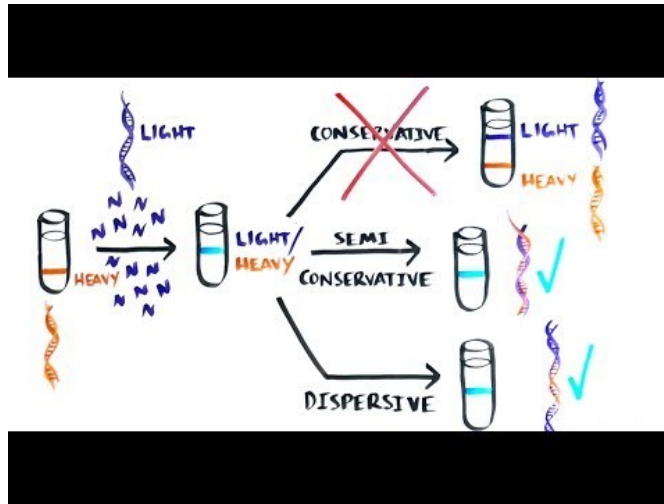
DNA Replication

During S phase of interphase, a cell replicates its entire genome in order to produce a new cell with an identical set of DNA.

- Each strand of DNA serves as a template for the creation of its **complementary strand**
- DNA replication is thus said to be ***semi-conservative***; that is, each new molecule of DNA contains one strand of the original DNA molecule and one newly synthesized strand

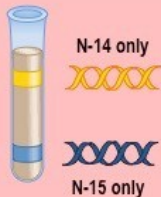
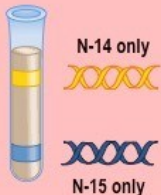


The Meselson-Stahl Experiment



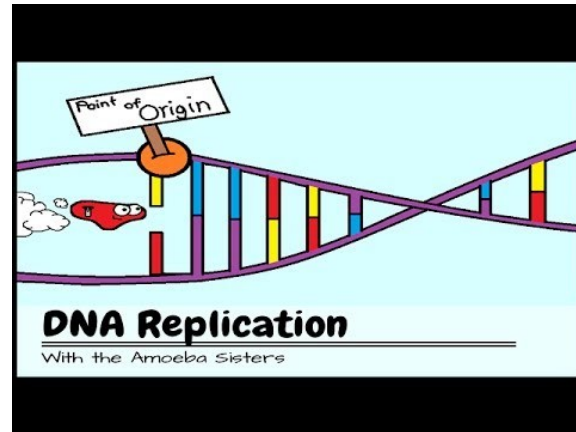
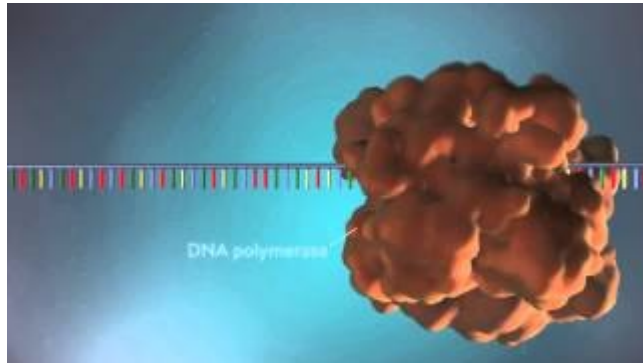
- The semiconservative nature of DNA replication was proven by the **Meselson-Stahl** Experiment.
- The semiconservative model maintains that the original double helix serves as a template for, but does not contribute to, a new double helix.

The Meselson-Stahl Experiment

ACTUAL OBSERVATIONS	PREDICTIONS		
	Conservative	Semiconservative	Dispersive
First Replication			
 N-15 / N-14	 N-14 only N-15 only	 N-15 / N-14	 N-15 / N-14
Second Replication			
 N-14 only N-15 / N-14	 N-14 only N-15 only	 N-14 only N-15 / N-14	 N-15 / N-14

DNA Replication

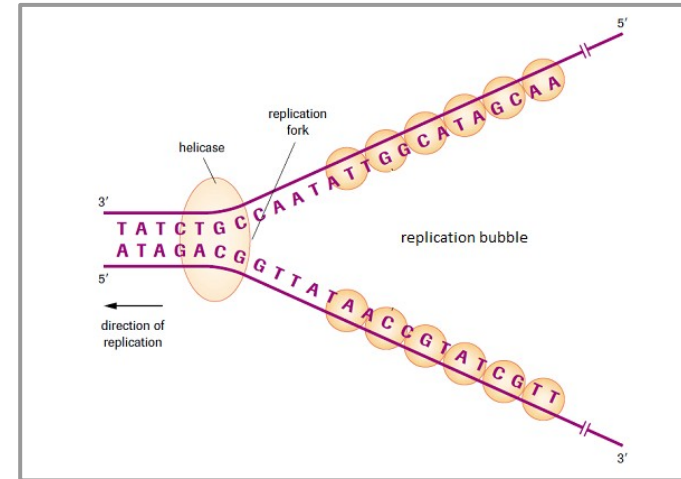
Watch these videos first, then read the summary in the next slides. You may then come back and re-watch these videos, or find others to better understand the process as a whole.



DNA Replication

STEP 1: INITIATION

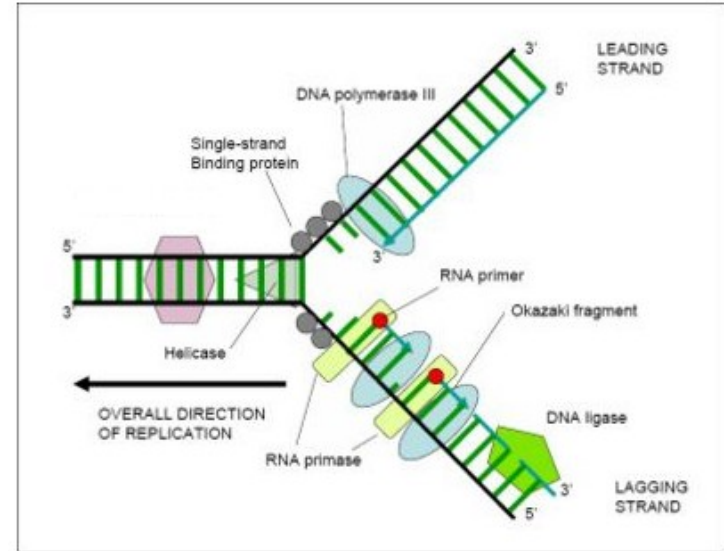
1. A group of enzymes known as *helicases* bind to a specific sequence of DNA referred to as the *replication origin*
2. The helicases then “cut” and unravel the double helix. The y-shaped area that forms where the DNA is split is known as the *replication fork*, while the unwound region is known as the *replication bubble*.
3. The single unwound strands of DNA serve as templates for replication



STEP 2: ELONGATION

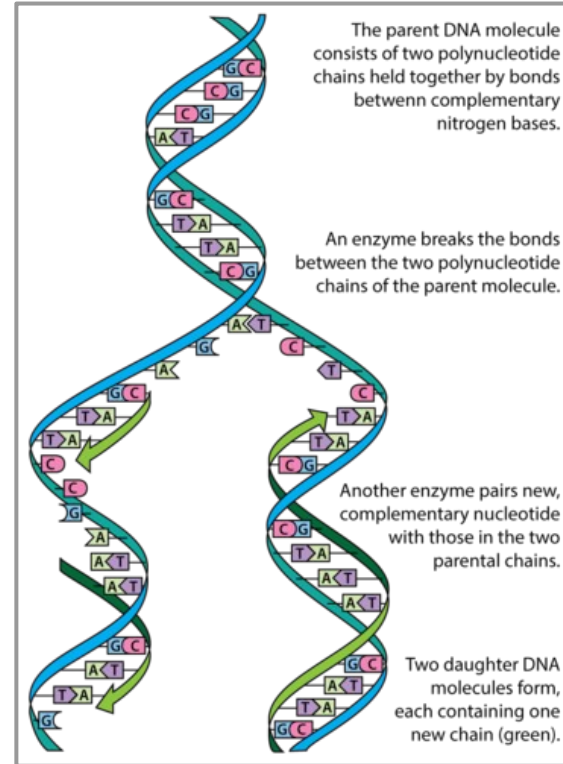
4. A **primer** sequence synthesized by the *RNA primase* enzyme signals **DNA polymerase** where to begin.
4. *DNA polymerase* begins to add nucleotides within the replication bubble, creating two new strands of DNA that are complementary to each existing strand.
4. Once replication is complete, the primers are removed by *DNA polymerase*
4. **Elongation always takes place in the 5' to 3' direction**, meaning that one strand will always be “**leading**” (replicated continuously) and the other will always be “**lagging**” (replicated in short segments known as *Okazaki fragments*)
4. Okazaki fragments of the lagging strand are spliced (‘glued’) together by an enzyme known as *DNA ligase*

Tip: There's often meaning in words in biology. Anything that ends in 'ase' is an enzyme, that 'makes something happen'. A polymer, is a long chain of monomer subunits, such as nucleotides. DNA polymerase, therefore is an enzyme that creates DNA polymerase.



STEP 3: TERMINATION

8. As soon as the newly formed strands are complete, they rewind automatically into their chemically stable helix structure
8. Replication proceeds until the two new DNA molecules separate; the dismantling of the replication complex is known as *termination*



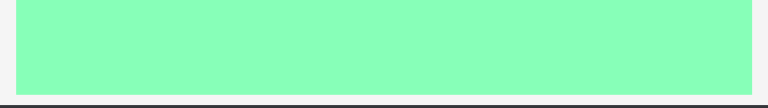
Key Enzymes to Know

ENZYME	FUNCTION
helicase	Cleaves (cuts) and unwinds DNA
RNA primase	Synthesizes primer to begin elongation process
DNA polymerase	adds new nucleotides from 5' end to 3' end of existing strand and removes primer sequences from DNA once replication is complete
DNA ligase	Splices together Okazaki fragments of lagging strand

Re-cap: DNA Replication

Note: Ignore any reference to DNA polymerase 3 vs 1 in the videos. These are only in prokaryotes, and we do not study them in this course. In eukaryotes we actually have more, but we will not distinguish between them and will refer to any as just DNA polymerase in this course.



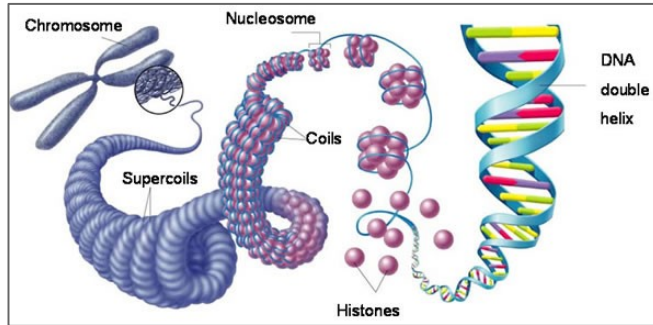


TOPIC 2: TRANSCRIPTION & TRANSLATION

- Properties of DNA vs. RNA
- Transcription of DNA into mRNA
- Translation of mRNA into an amino acid sequence
- Role of tRNA in translation
- Role of rRNA in translation
- Codons vs. anticodons
- Introns vs. exons

Textbook pgs. 636-642

Gene Expression

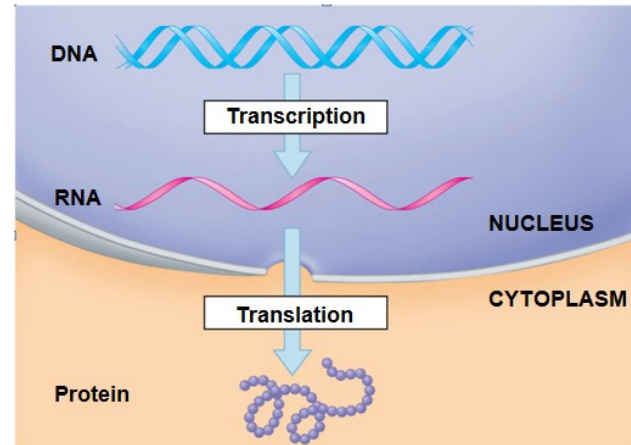


- Recall that DNA provides the “instructions” for each of our cells.
- Each segment of DNA (gene) is responsible for encoding a specific protein, which carries out a specific function.
- Not every gene is expressed in each cell. This is what allows cells to specialize and function differently.
- Some genes do not appear to code for any proteins... “junk” DNA?

Central Dogma of Biology

In order for a segment of DNA to be expressed by a cell, it must first be transcribed into mRNA (messenger RNA) and then translated into a sequence of amino acids (a protein). This is referred to as the “**central dogma**” of gene expression.

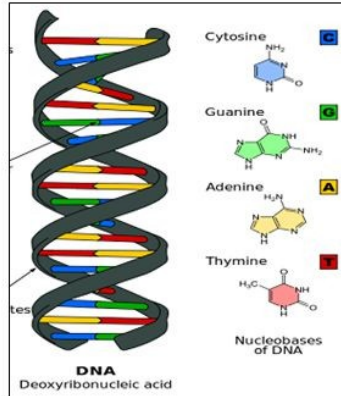
- Allows different cells to express different genes
- So, although all cells contain the same DNA, not all cells behave the same



Okay, so what is RNA exactly?

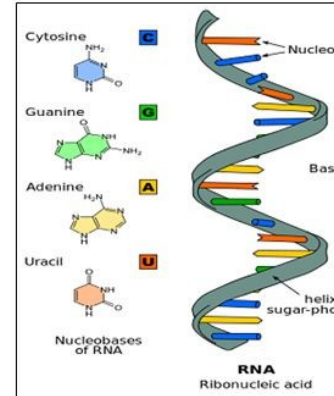
DNA

- Has deoxyribose sugar
- Has two strands (double-helix)
- Has C, G, A, and T bases



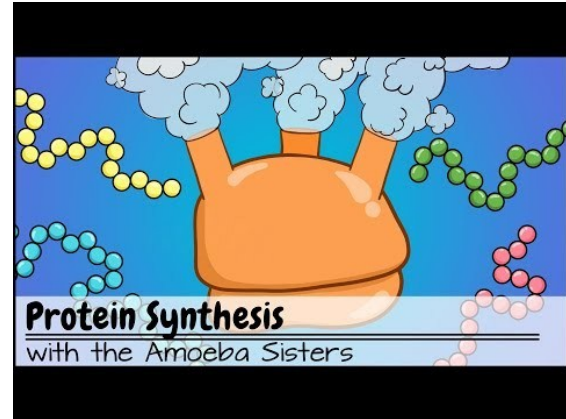
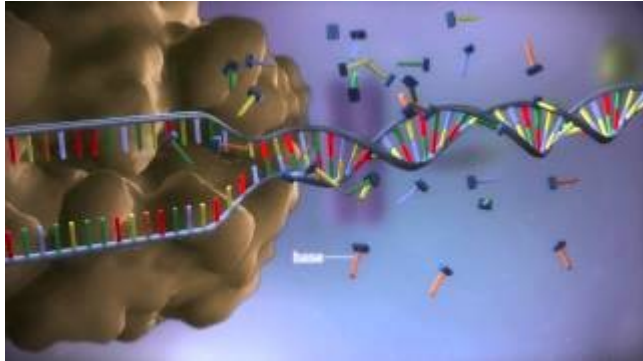
RNA

- Has ribose sugar
- Has one strand
- Has C, G, A and U (Uracil) bases



From DNA to Protein

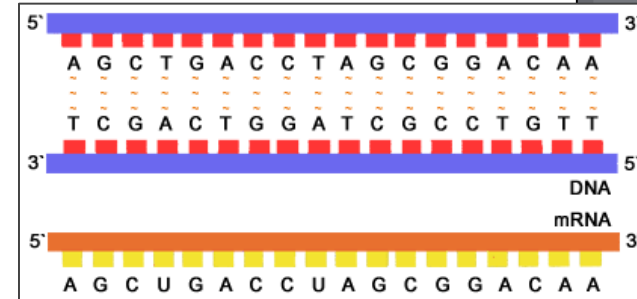
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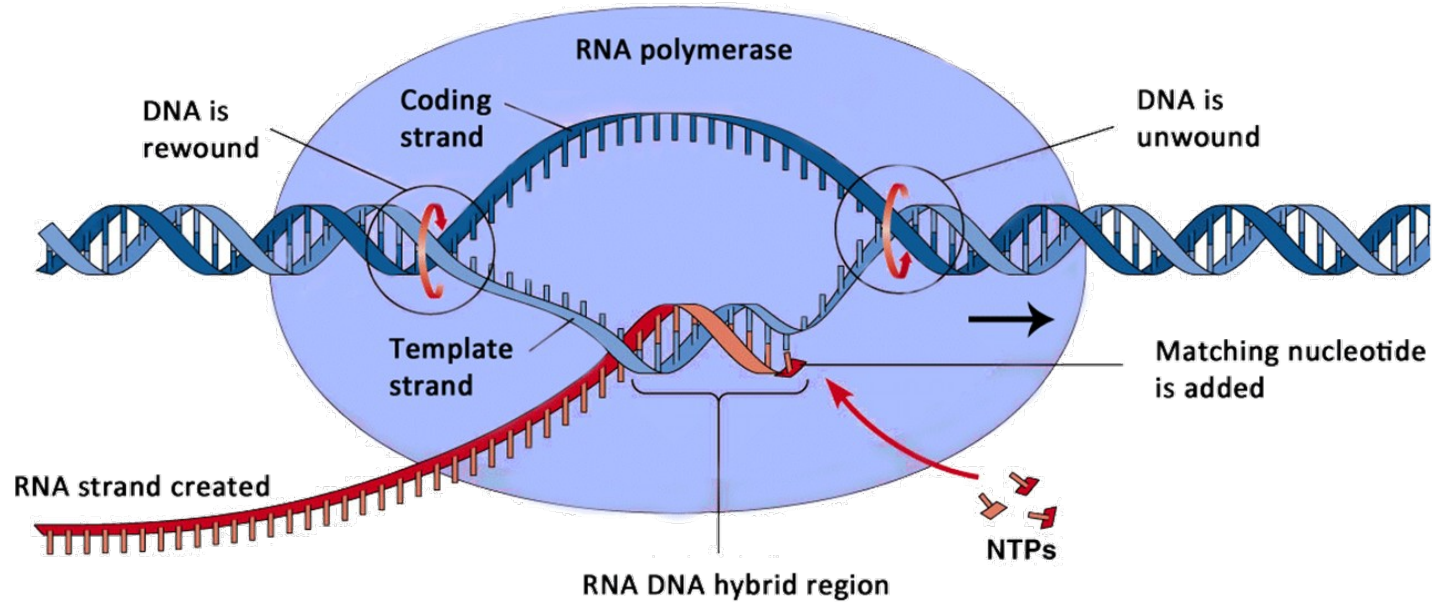
1) Transcription

During transcription, the information in a segment of double-stranded DNA is copied into a single strand of messenger RNA (mRNA)

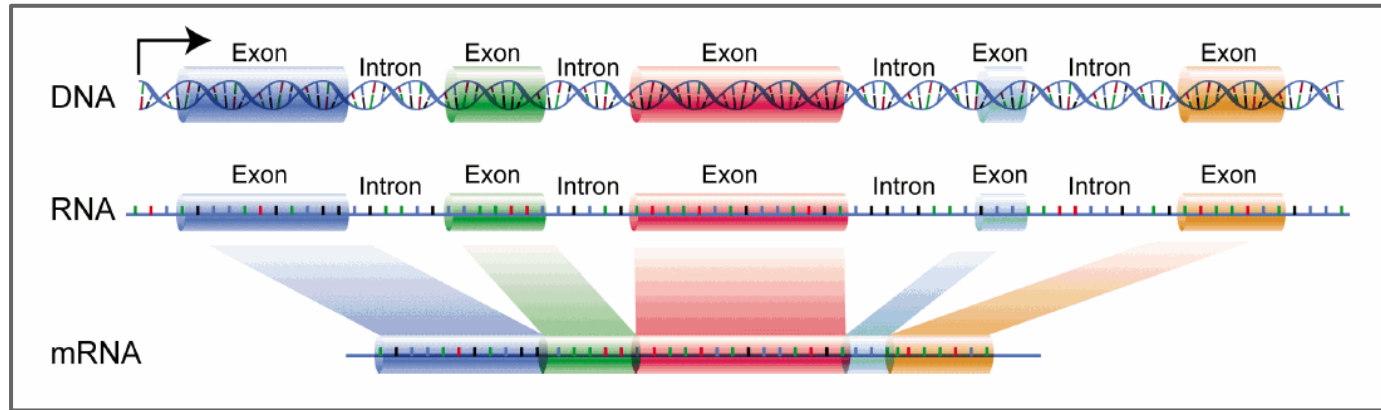
- Transcription takes place within the nucleus of a cell.
- Only one strand of the DNA (the **coding or “sense”** strand) is transcribed into mRNA. Since only one strand is transcribed, there is no need for Okazaki fragments here.
 - The other strand is referred to as non-coding or antisense strand.
- Transcription is initiated when **RNA polymerase binds to the promoter region** of the coding strand and begins to add complementary base pairs in the 5' to 3' direction.
- Whenever a thymine should be inserted, it is replaced by uracil (U's instead of T's) because in RNA, there is no thymine.
- Once RNA polymerase reaches a **termination sequence**, the mRNA strand is complete and can leave the nucleus to be translated into a protein.



1) Transcription



Extra: mRNA processing

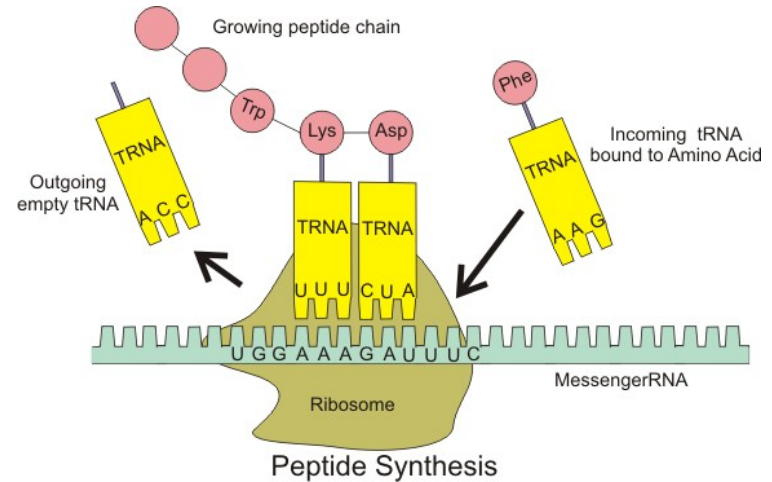


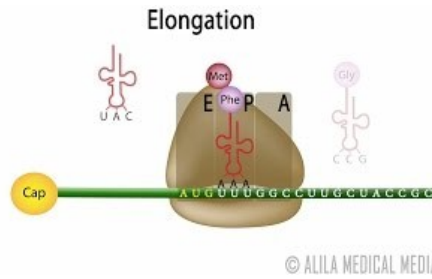
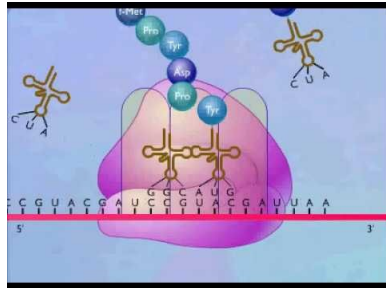
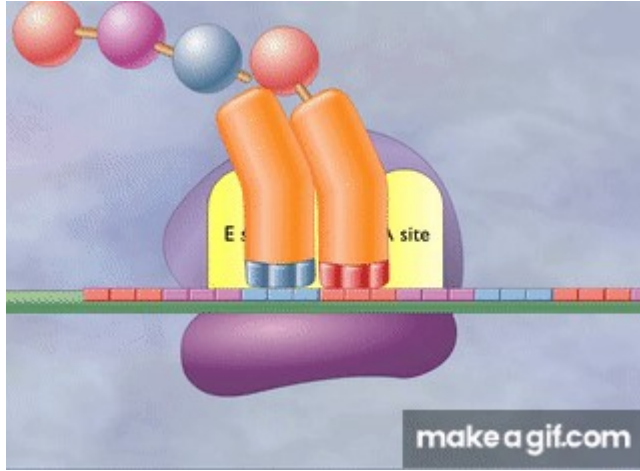
- Before translation takes place, the mRNA sequence is modified to remove sequences of mRNA called *introns*.
- Introns are sequences of RNA that do not code for proteins.
- The remaining RNA sequences are referred to as *exons*.
- This modification process occurs only in eukaryotes.

2) Translation

Once DNA is transcribed into mRNA, it then leaves the nucleus and moves into the cytoplasm of the cell to be translated into a protein.

- The mRNA sequence is split into **codons** (groups of three base pairs), each of which links to a specific amino acid through the use of transfer RNA (tRNA).
- Codons are linked to amino acids by **tRNA anticodons**.
- Ribosomes (ribosomal RNA - rRNA) serve as the functional units of protein synthesis, working with tRNA to translate mRNA into amino acids





- Ribosomes are typically free-floating within the cytoplasm; these ribosomes produce proteins for use within the cell.
- Ribosomes consist of two subunits: a large subunit and a small subunit. The mRNA sequence is “sandwiched” between these two subunits when it is translated.
- Each codon to be translated arrives at the “A” site of the ribosome with the help of tRNA. It then moves to the “P” site, where the correct amino acid is attached to the growing polypeptide chain. It then exits at the “E” site.
- Bound ribosomes produce proteins for export.

Codons to Amino Acids

- Table of mRNA codons (in your data booklet) and corresponding amino acids
- For example, if the **mRNA** sequence is AGC, the amino acid 'serine' will be inserted during translation
- Remember!! You must be using mRNA to look up the amino acid in this table.

AUG - is an **initiator or start codon**. The ribosomes will 'scan' the mRNA sequence, and only begin translating at AUG. AUG therefore also set the reading frame for the mRNA.

UAA, UAG, and UGA - **terminator or stop codons**. The ribosome will stop translating here and the protein is finished.

Messenger RNA Codons and Their Corresponding Amino Acids

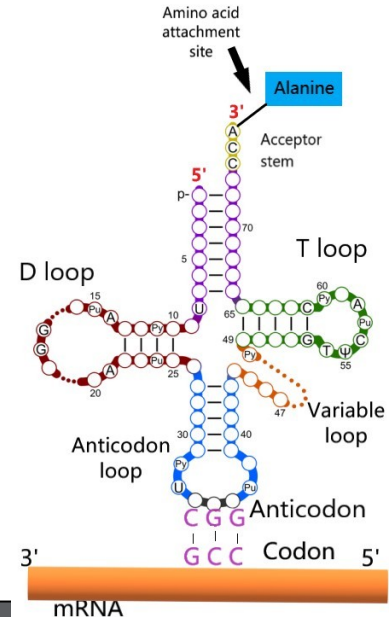
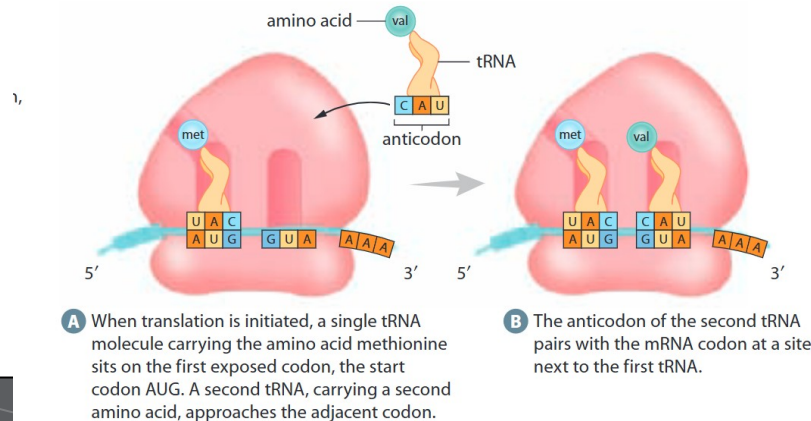
First Base	Second Base				Third Base
	U	C	A	G	
U	UUU phenylalanine	UCU serine	UAU tyrosine	UGU cysteine	U
	UUC phenylalanine	UCC serine	UAC tyrosine	UGC cysteine	C
	UUA leucine	UCA serine	UAA stop**	UGA stop**	A
	UUG leucine	UCG serine	UAG stop**	UGG tryptophan	G
C	CUU leucine	CCU proline	CAU histidine	CGU arginine	U
	CUC leucine	CCC proline	CAC histidine	CGC arginine	C
	CUA leucine	CCA proline	CAA glutamine	CGA arginine	A
	CUG leucine	CCG proline	CAG glutamine	CGG arginine	G
A	AUU isoleucine	ACU threonine	AAU asparagine	AGU serine	U
	AUC isoleucine	ACC threonine	AAC asparagine	AGC serine	C
	AUA isoleucine	ACA threonine	AAA lysine	AGA arginine	A
	AUG methionine*	ACG threonine	AAG lysine	AGG arginine	G
G	GUU valine	GCU alanine	GAU aspartate	GGU glycine	U
	GUC valine	GCC alanine	GAC aspartate	GGC glycine	C
	GUA valine	GCA alanine	GAA glutamate	GGA glycine	A
	GUG valine	GCG alanine	GAG glutamate	GGG glycine	G

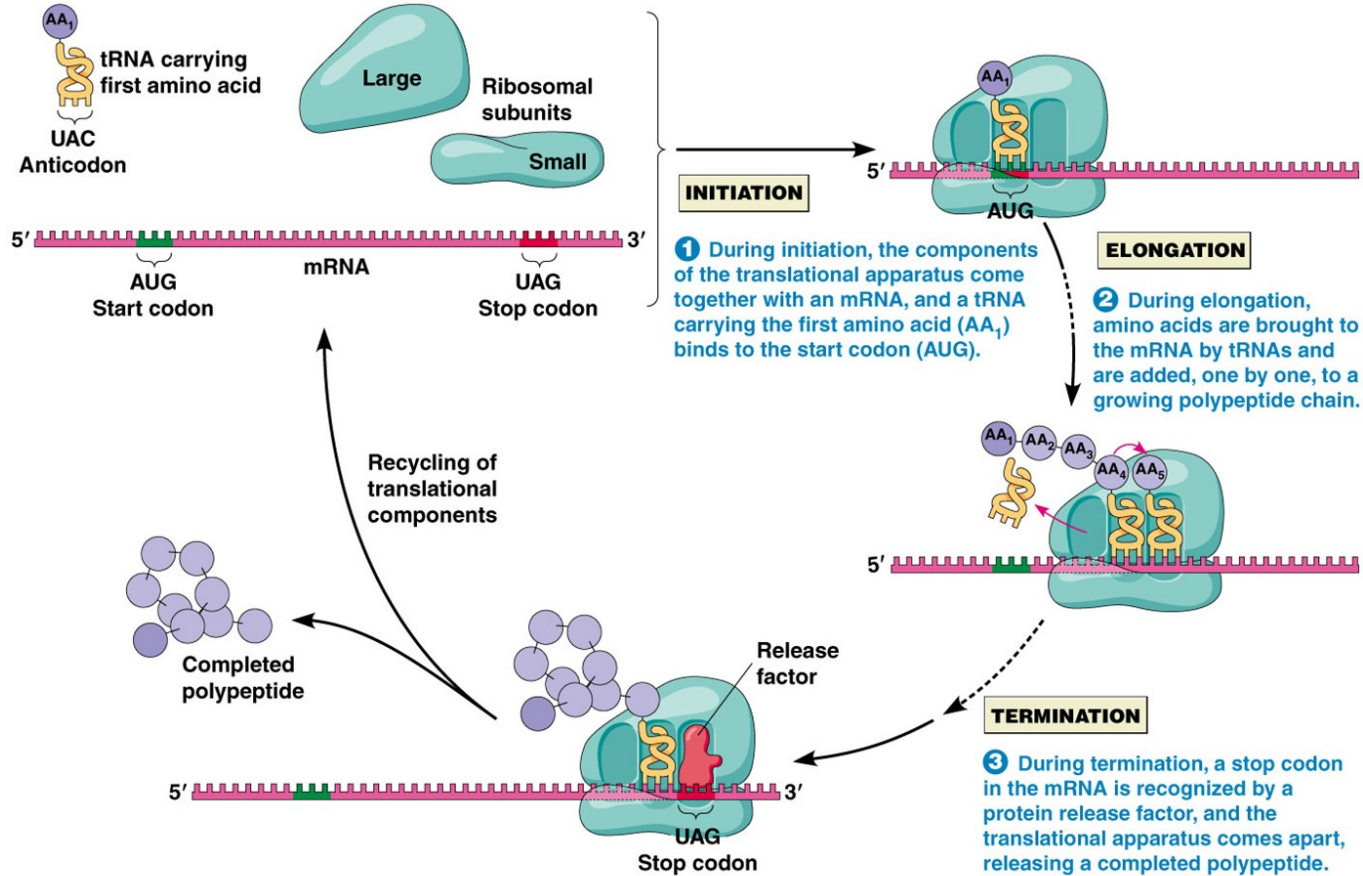
*Note: AUG is an initiator codon and also codes for the amino acid methionine.

**Note: UAA, UAG, and UGA are terminator codons.

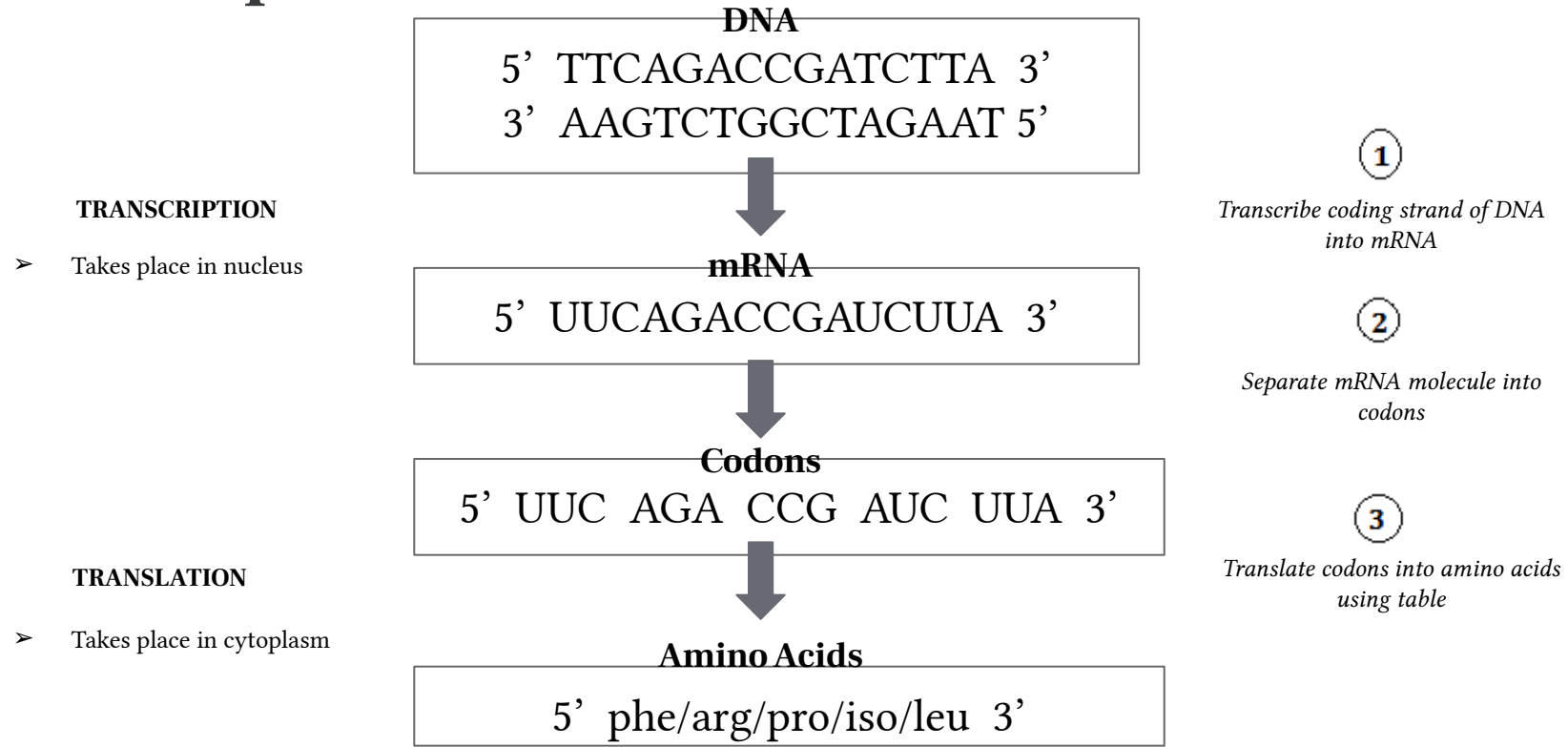
A closer look at tRNA

- Although tRNA is still single stranded, like other RNA molecules, it also has a looped structure.
- We don't need to worry too much about the loops, but one of its loops contains the **anticodon**, through which the tRNA molecule **binds to the complementary codon in the mRNA** molecule.
- E.g. a tRNA with an anticodon of CGG, will carry alanine, because the complementary mRNA codon is GCC. REMEMBER: You have to look up mRNA in your data booklet.





Example:



Transcription and Translation Recap

TRANSCRIPTION & TRANSLATION

mRNA codon table:

	U	C	A	G
U	UUU → Phe UUC → Phe UUA → Leu UUG → Leu	UUU → Phe UUC → Phe UUA → Leu UUG → Leu	UUU → Phe UUC → Phe UUA → Leu UUG → Leu	UUU → Phe UUC → Phe UUA → Leu UUG → Leu
C	CUU → Leu CUC → Leu CUA → Leu CUG → Leu	CUU → Leu CUC → Leu CUA → Leu CUG → Leu	CUU → Leu CUC → Leu CUA → Leu CUG → Leu	CUU → Leu CUC → Leu CUA → Leu CUG → Leu
A	AUU → Ile AUC → Ile AUA → Ile AUG → Met	AUU → Ile AUC → Ile AUA → Ile AUG → Met	AUU → Ile AUC → Ile AUA → Ile AUG → Met	AUU → Ile AUC → Ile AUA → Ile AUG → Met
G	GUU → Val GUC → Val GUA → Val GUG → Val	GUU → Val GUC → Val GUA → Val GUG → Val	GUU → Val GUC → Val GUA → Val GUG → Val	GUU → Val GUC → Val GUA → Val GUG → Val

DNA: 5' ATGATTTTATTTTAA 3' (coding)
 3' TACAAAATACCAAT 5' (template)

mRNA: 5' AUG 3'

protein: Met

Practice #1

- Transcribe the following strand of DNA into mRNA:

DNA: GCT TAG GCC CAT CTA

mRNA: CGA AUC CGG GUA GAU

- Translate your mRNA strand above into a sequence of amino acids:

mRNA: CGA AUC CGG AUA GAU

AA sequence: arginine-isoleucine-arginine-valine-aspartate

Practice #2

- How many nucleotides are required to code for the following sequence of amino acids?

Leu-Tyr-Arg-Trp-Ser

15 nucleotide (each AA requires a codon, which is a nucleotide triplet)

- Is it possible to determine the mRNA sequence that is responsible for producing the sequence of amino acids above? Explain...

No. More than one codon codes for most AAs. For example, leucine can come from an mRNA sequence of UUA, UUG, or anything that begins with CU_

Redundancy

- The previous practice question illustrates the **redundancy** of the genetic code. This means that most amino acids are encoded for by different mRNA codons, and therefore different DNA nucleotide triples.
 - But no codon will ever specify for two different amino acids
- Redundancy is a good thing, when we consider mutations. For example, if the normal DNA sequence is CAA, but during DNA replication DNA polymerase inserts a G instead of the second A and now the DNA sequence is CAG, it still codes for glutamine. The finished protein will be no different.

Practice #3

- Identify a **single-letter** mutation that could occur in the following DNA sequence to produce a stop codon at some point in the amino acid:

AACGAATATCGG

Remember, that AAs are determined based on mRNA codons. The STOP codons are UAA, UAG, and UGA, which would correspond to DNA sequences of ATT, ATC, and ACT.

AAC GAA TAT CGG → ATC GAA TAT CGG

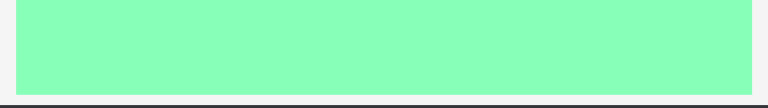
Practice #4

Complete the following table:

Coding strand	ACGGCTTTACAC
Non-coding strand	TGCCGAAATGTG
mRNA	UGCCGAAAUGUG
tRNA	ACGGCUUUACAC
AA Sequence	cys-arg-asp-val

Re-Cap: Types of Nucleic Acids

NUCLEIC ACID	STRUCTURE	FUNCTION
DNA	Double helix	Stores genetic info
mRNA	single strand	Carries genetic info from DNA in nucleus to ribosomes in cytoplasm for protein synthesis. Transcribed from DNA.
tRNA	Single strand	Carries amino acids to the correct codon site during protein synthesis. Complementary to mRNA
rRNA	Single strand	Combines to form a ribosome, the main structure required for protein synthesis



TOPIC 3: MUTATIONS AND GENETIC TECHNOLOGIES

- Causes of mutations: Point mutation vs. frameshift mutation
- Effects of mutations: Silent vs. missense vs. nonsense mutation
- Mitochondrial DNA
- Recombinant DNA
- Gel electrophoresis
- Taq polymerase & PCR

Textbook pgs. 643-662

Mutations

Mistakes in the coding of genetic information often take place during DNA replication, transcription and/or translation. Most of the mistakes get corrected right away, by various proofing mechanisms cells have. Mistakes that do not get corrected, however, result in the production of a different sequence than what was intended. These mistakes are referred to as *mutations*. Certain types of mutations have more severe consequences than others.

- Mutations that occur in somatic cells (body cells) are not passed on to offspring (e.g. cancer)
- Mutations that occur in the germ line (reproductive cells) are passed on to offspring (e.g. trisomy 21)



Types of Mutations

1) POINT MUTATION

- Chemical change that affects just one nucleotide
- This involves the substitution of one nucleotide for another (e.g. a “G” instead of an “A”)
- Point mutations can be
 - **silent:** no change in AA
 - **missense:** change to new AA
 - **nonsense:** change to stop codon

CAG UUA CCC



glu/leu/pro

CAG UUG CCC



glu/leu/pro

Mutation changes “A” to “G” in second codon, but does not result in a different amino acid

Point Mutations

Silent

CAG UUA CCC



glu/leu/pro

CAG UUG CCC



glu/leu/pro

Mutation changes “A” to “G” in second codon, but does not result in a different amino acid. Correct protein is made.

Missense

CAG UUA CCC



glu/leu/pro

CAG UCA CCC



glu/ser/pro

Mutation changes “U” to “C” in second codon, which results in different AA. This can have varying impacts, depending on how similar the AAs are and where in the protein it is.

Nonsense

CAG UUA CCC



glu/leu/pro

CAG UAA CCC



glu/STOP

Mutation changes “U” to “A” in second codon, which results in a STOP codon. This means that no functional protein is made

Causes of Mutations

2) FRAMESHIFT MUTATION

- An **insertion** or **deletion** of a nucleotide that causes the **entire reading frame of the gene to be altered**
- Typically has more severe consequences, as it changes the protein entirely (not just a single amino acid)

CAG UUA CCC



glu/leu/pro

CAGG UUA CCC



CAG GUU ACC C



glu/val/thr

Practice

Identify the type of mutation in each of the following situations

- a. Original strand: AUG UUU UUG CCU UAU CAU CGU
Mutated strand: AUG UUU UUG CCU UAC CAU CGU

Silent mutation

- a. Original strand: AUG UUU UUG CCU UAU CAU CGU
Mutated strand: AUG UUU UUG CCU UAA CAU CGU

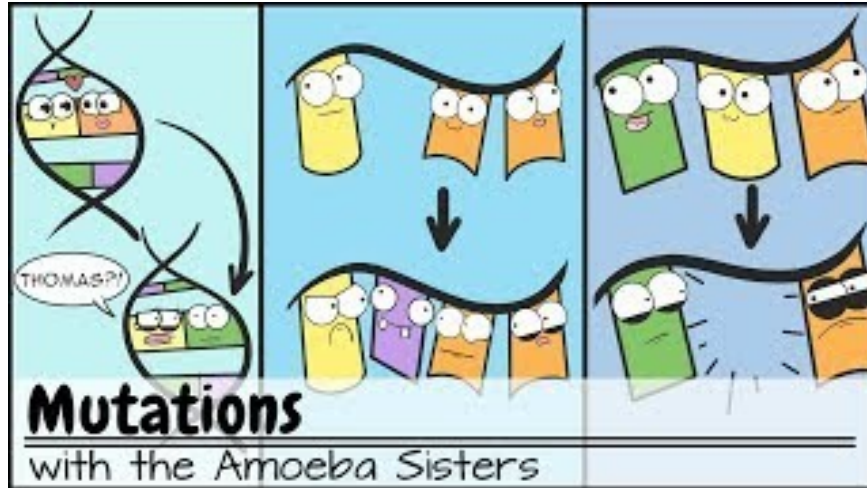
Nonsense mutation

- a. Original strand: AUG UUU UUG CCU UAU CAU CGU
Mutated strand: AUG UUU UUG CCU AUC AUC GU

Deletion frameshift

Remember, you always need to be considering the mRNA sequence. So if a similar question gave you DNA, you would first have to transcribe it into mRNA to identify the type of mutation.

Mutations Summary



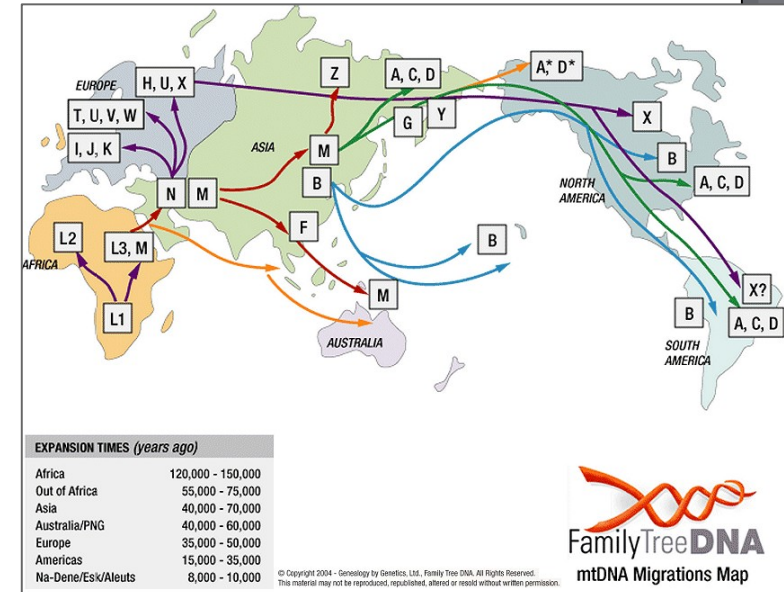
Mutations & Natural Selection



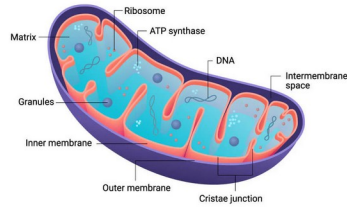
Mitochondrial DNA

According to our most current understanding, some mitochondria may be donated by the sperm and therefore mtDNA can be paternal, but this is much less frequent and we will ignore for Bio 30.

- DNA is not just found in the nucleus of the cell; it is also stored in the mitochondria. However, unlike nuclear DNA, mtDNA is only inherited from the mother. This is because all cytoplasm and organelles, such as mitochondria, come from the egg; sperm just contributes DNA.
- By analyzing the mtDNA of a species, we can actually trace back lineages through time to identify a common ancestor
 - mtDNA mutations usually aren't lethal, so mutations can accumulate in great numbers over time
 - mtDNA is also shorter, which makes sequencing easier
 - The more similar mtDNA sequences, the more closely related organisms are



MITOCHONDRIA

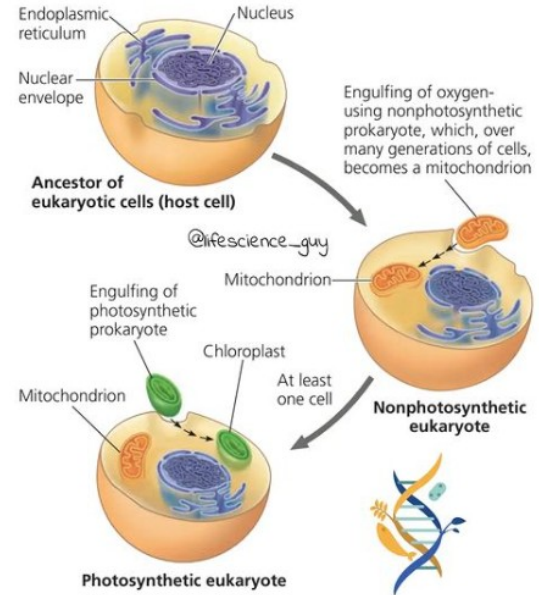


Why do mitochondria contain DNA?

- Mitochondria (and chloroplasts) have their own DNA, because they used to be their own free living bacteria. Through the process of endosymbiosis they came to be part of eukaryotic cells.



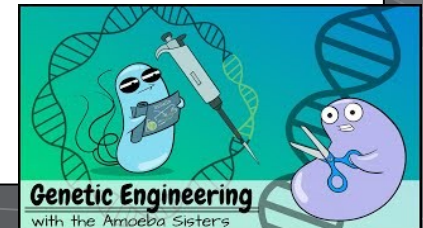
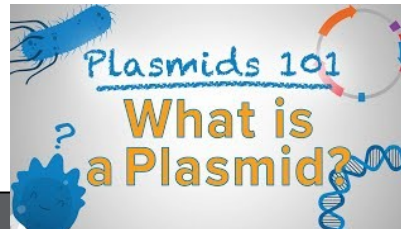
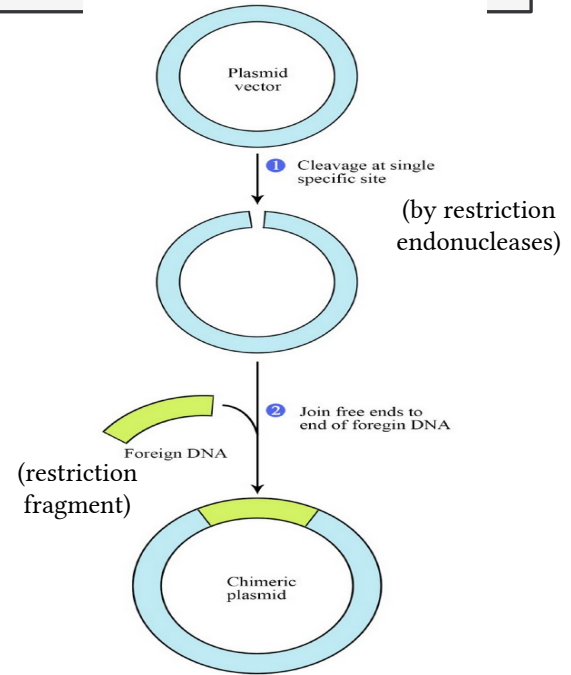
The Endosymbiont Theory



Genetic Engineering

A molecule of DNA that includes genetic material from different sources is called *recombinant DNA*. It is produced through the process of *genetic engineering*.

- Genetic engineering is used globally to produce organisms with desirable traits that could not be produced via selective breeding (Genetically Modified Organisms - GMOs).
 - E.g. herbicide resistant canola or corn
 - E.g. mass-production of human insulin by bacteria.
- Genetic engineering often involves taking **plasmids** from bacteria (circular segments of DNA which contain their genetic info) and re-programming them to express certain genes.
- **Restriction endonucleases (restriction enzymes)** “cut” the DNA sequence at a specific recognition site to allow for the insertion of a desired gene, which is “glued” into the genome using DNA ligase.

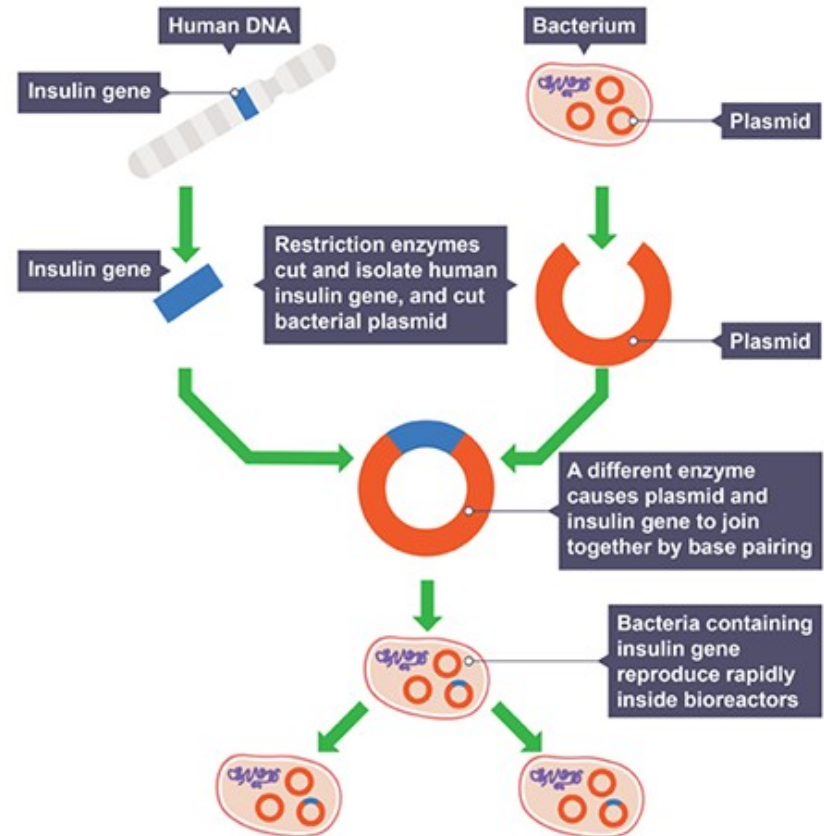
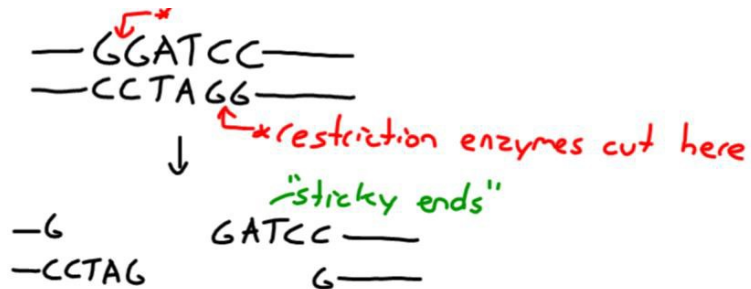


Genetic Engineering

General steps involved with genetic engineering and creating insulin producing bacteria:

1. **Use restriction enzymes to cut and remove functioning insulin gene** from a human pancreatic cell
2. **Use the same restriction enzyme on a plasmid** to get same 'sticky ends'. These are usually palindromic sequences.

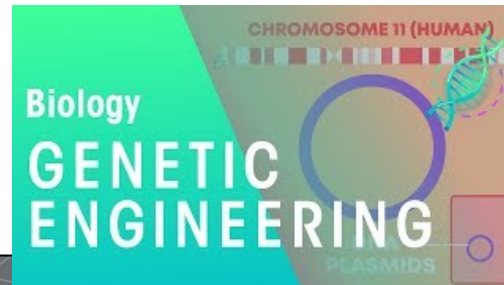
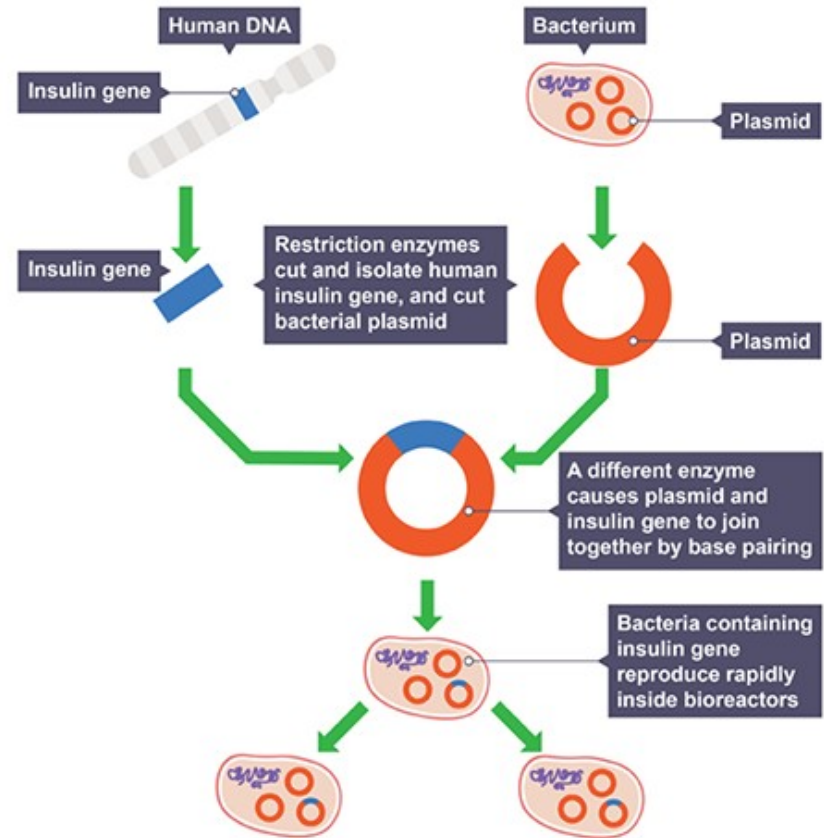
For example:

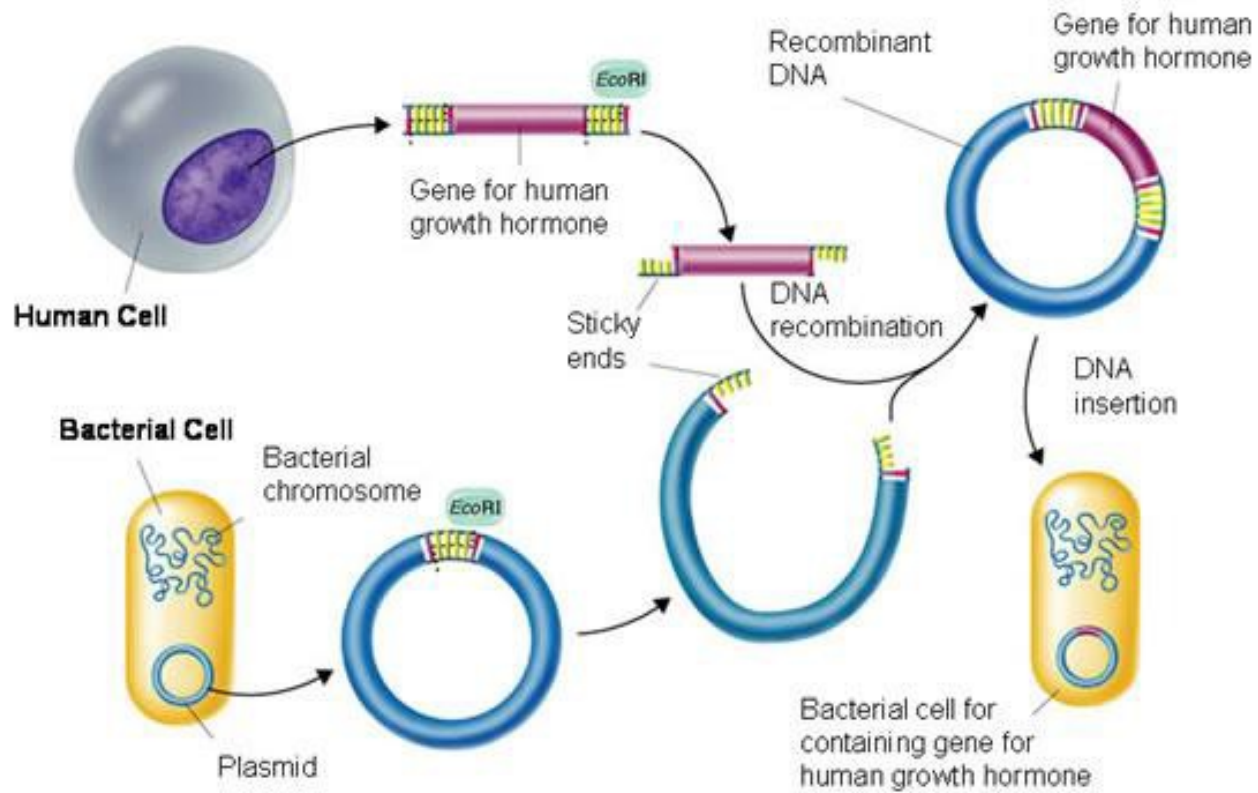


Genetic Engineering

Cont.

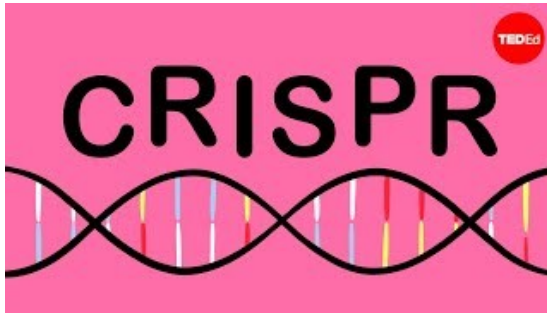
3. Mix the samples and use **DNA ligase to fuse/glue** the cut-out insulin gene into the plasmid
3. Allow bacteria to take up plasmid from their environment by transformation
3. Allow growth of bacterial colonies on petri dish
3. Isolate insulin producing bacteria and extract insulin





Extra: Genetic Engineering

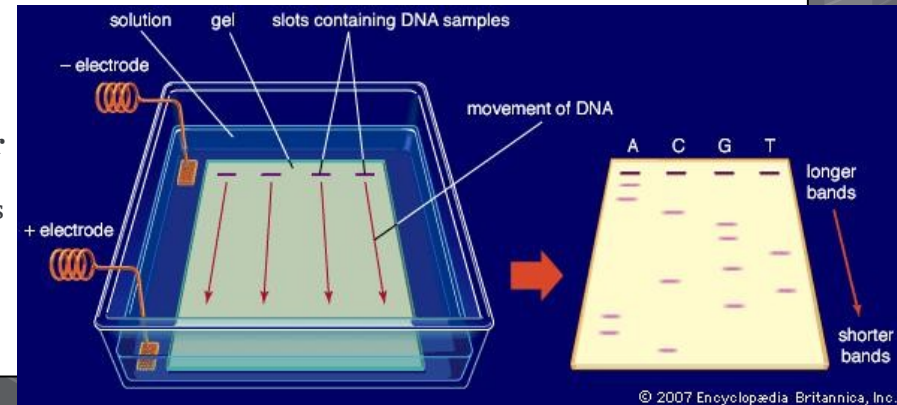
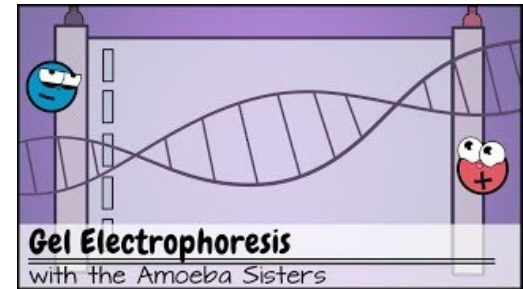
- Genetic engineering is a fast growing field of science. There are so many fascinating and instructional videos on the internet. I encourage you to seek out some videos on this topic to learn more. Here are a few to get started with:

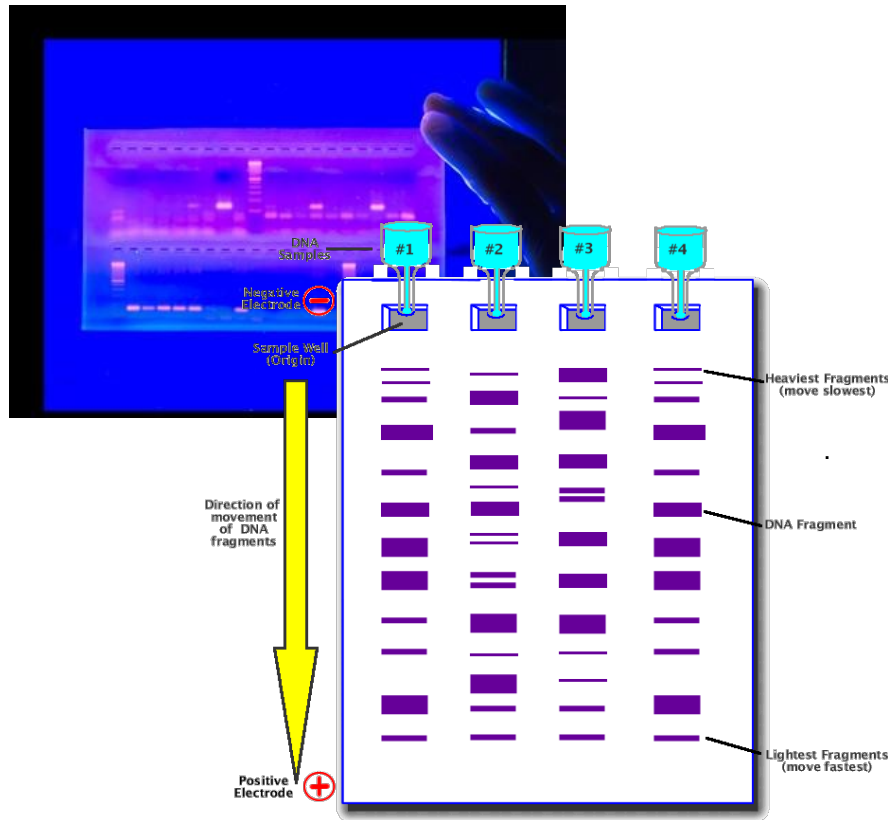


Gel Electrophoresis

Through a process known as *gel electrophoresis*, we can analyze and compare the DNA of any organism:

- Involves applying a current to a gel made of agarose sugar
→ **there is a positive electrode end and a negative electrode end**
- **Restriction enzymes** cut DNA at specific points, forming a series of varying sized fragments
- **DNA is negatively charged**, so it will migrate towards the positive electrode
- The smaller the fragment, the easier it is to move through the gel, therefore **smaller fragments will migrate further**
- The banding pattern of the gel allows us to compare samples of DNA

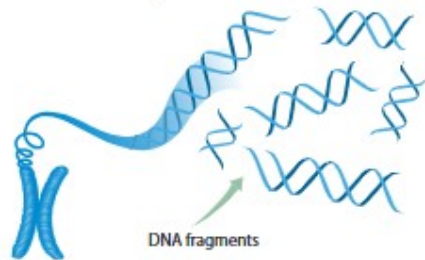




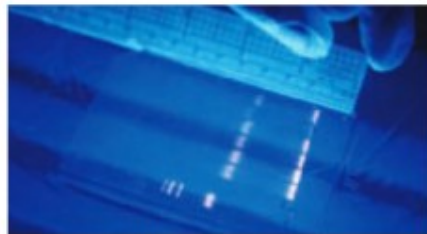
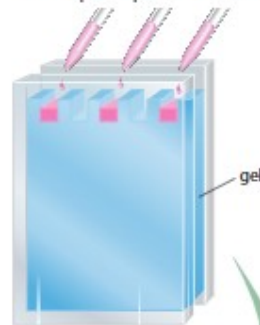
Gel Electrophoresis
(Creating a DNA Profile)

- E.g. Samples #1 and #4 display the same banding pattern, indicating that the organisms they were taken from are closely related, most likely being the same individual

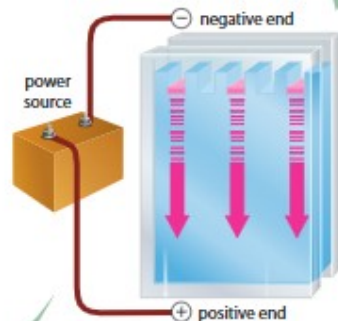
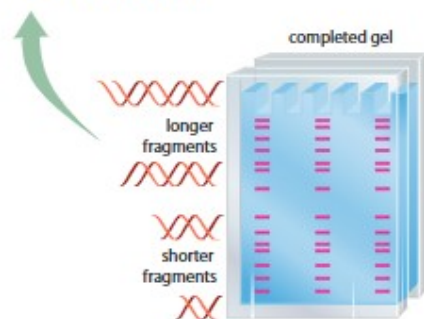
A Restriction enzymes Either one or several restriction enzymes are added to a sample of DNA. The enzymes cut the DNA into fragments.



B The gel A gel, with a consistency similar to gelatin, is formed so small wells are left at one end. Small amounts of the DNA sample are placed into these wells.



E Before the DNA fragments are added to the wells, they are treated with a dye that glows under ultraviolet light, allowing the bands to be studied.

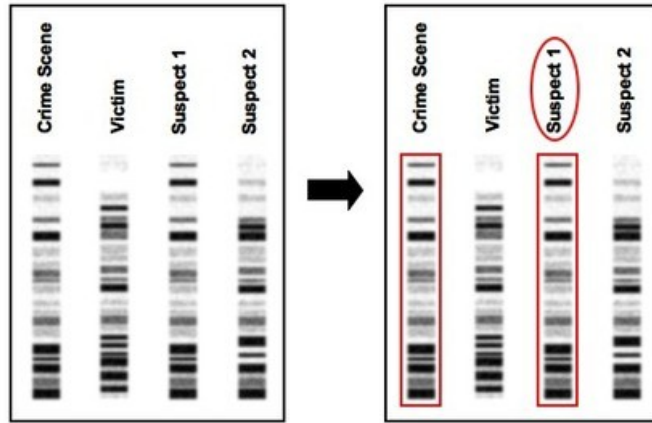


C The electrical field The gel is placed in a solution, and an electrical field is set up so one end of the gel is positive and the other end is negative.

D The fragments move The negatively charged DNA fragments travel toward the positive end. The smaller the fragment, the faster it moves through the gel. Fragments that are the farthest from the well are the smallest.

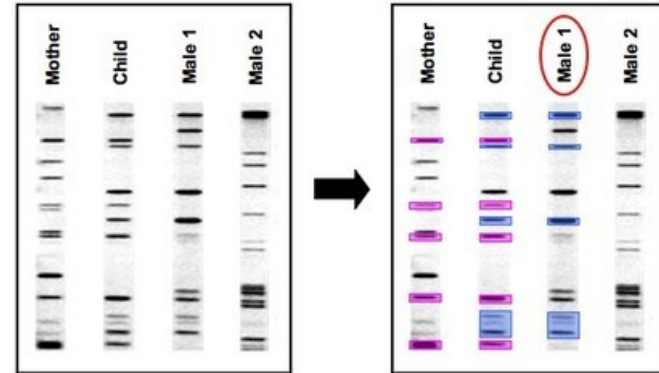
Applications of Gel Electrophoresis

Who committed the crime?



Note: To identify a suspect, it needs to be a perfect match. Every band has to have an exact match.

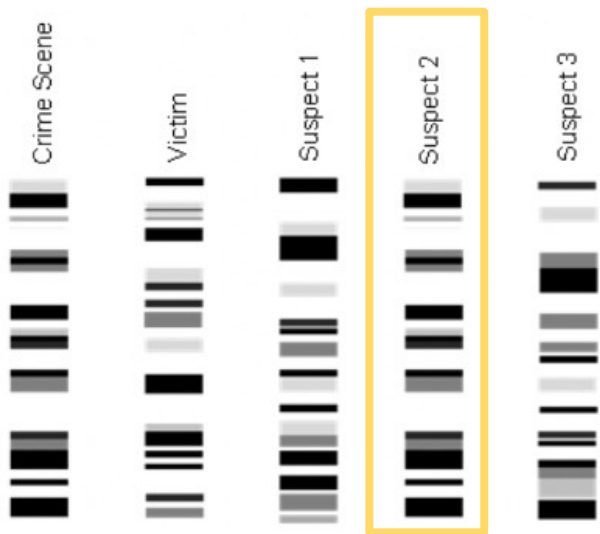
Who is the father?



Note: Remember that a child has half of its DNA from mom and half from dad. To identify a father, every band that could not from the mother, has to be from the father.

Practice

Who committed the crime?



Who is the father?

